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Chapter 6. VPN privacy: Aura, Mullvad, and your options

The payoff of this chapter: understand what a VPN actually does for you, see how a mainstream consumer VPN (I started with Aura) compares to a privacy-focused option (Mullvad), and learn the Tailscale-native way to route traffic, the *exit node*, so you can pick the setup that fits. The messier question of how all this behaves *away from home* (and how it collides with your Pi-hole) gets its own treatment in [Chapter 7](#).

Everything so far has been about reaching *into* your homelab. This part is the opposite direction: making your devices' traffic leave through a VPN, so the websites and apps you use can't see your real IP address, and so your ISP or a café's Wi-Fi can't see what you're doing.

Maybe you already run a consumer VPN to change locations. I started with **Aura VPN**, and it's a perfectly good one. The goal here isn't to talk you out of whatever you run, it's to lay out the landscape clearly: what a VPN does, where a consumer app like Aura fits, where **Mullvad** fits, and the one Tailscale concept (the **exit node**) that ties VPNs into the homelab you've been building.

What a VPN does (and what it doesn't)

A VPN routes all your internet traffic through a remote server before it reaches the wider internet. Two practical consequences:

- **Your IP is hidden.** Sites see the VPN server's address, not yours, useful for privacy and for appearing to be in another location.
- **Your local network can't snoop.** On untrusted Wi-Fi, the airport or hotel can see only that you're talking to a VPN, not what you're doing.

What a VPN does *not* do: it isn't anonymity (the VPN provider can see your traffic, so provider trust matters), and it doesn't block ads by itself unless the provider offers DNS-level filtering. Keep that ad-blocking caveat in mind, it's the seam where VPNs and your Pi-hole rub against each other, which is exactly the Chapter 7 discussion.

The Tailscale exit node: a VPN built from your own tailnet

Before comparing products, here's the concept that makes Tailscale relevant to VPNs at all.

Normal Tailscale is an **overlay network**: it only carries traffic *between* your own devices (laptop homelab phone). Your ordinary internet traffic still leaves through your local connection; Tailscale doesn't touch it.

An **exit node** changes that. You designate one device on your tailnet as “the exit,” and other devices route **all** their internet traffic through it, the full default route, not just tailnet traffic. To the outside world, your traffic now appears to come from the exit node's IP. That's exactly how a traditional full-tunnel VPN behaves, except the “VPN server” is a node on your own tailnet. Only one exit node is active per device at a time, and you switch it off by selecting “None.”

This matters because Tailscale can use **Mullvad's servers as exit nodes**, giving you a real privacy VPN *inside* the same Tailscale app you already run, with no second VPN client. That's the key to making VPN and homelab coexist, and it's why Mullvad gets special attention below.

Your options (three simple, one advanced)

Option A: Tailscale's Mullvad add-on (*best fit for this homelab*)

Tailscale sells access to Mullvad's worldwide WireGuard servers as a paid add-on **purchased through Tailscale**, no separate Mullvad subscription. Mullvad's servers simply appear as selectable exit nodes inside your tailnet.

- **Cost:** about **\$5/month**. Tailscale describes this as 5 “licenses,” where one license covers up to 5 devices. (Tailscale's marketing page and its docs state the device math slightly differently, confirm the exact count at the checkout screen before buying.)
- **Enable it:** admin console → **Settings** → the **Mullvad** section → **Configure**, complete checkout, then grant your devices access.
- **Use it (Linux):**

```
tailscale exit-node list           # list Mullvad cities/servers
tailscale set --exit-node=<mullvad-node> # route everything through it
tailscale set --exit-node=         # turn it back off
```

On iOS / macOS / Android: the **...** menu → **Use exit node** → pick a Mullvad location.

- **Why it's the best fit here:** it *is* Tailscale, so it coexists with everything you've built, and crucially, it's the only option that works on iPhone *without* fighting Tailscale (more on that in Chapter 7). The `--exit-node` choice is persistent across reboots, so “always on” is just setting it once.

Option B: A standalone VPN app (Aura, or the Mullvad app)

This is the familiar consumer model, the one I started on with **Aura**: a self-contained VPN app on each device that you toggle on to change location.

- **Aura**: part of the Aura identity-protection suite. A closed consumer app, easy to use, location-switching, with split tunneling on its supported platforms. It's fine for casual "make me look like I'm elsewhere." Its limitations for *this* project: it's app-only (there's no client you can run on the headless Pi, and no WireGuard config files to reuse), and like any full-device VPN it takes over DNS while it's on.
- **Mullvad app**: a flat **€5/month** (~\$5.40), up to 5 devices, includes a proper **kill switch**, and (unlike Aura) Mullvad hands you **WireGuard config files** you can run anywhere, including on the Pi. That openness is what makes the advanced setups in Chapter 7 possible.

The catch with *any* standalone VPN app is how it coexists with Tailscale, a genuinely thorny topic that's the heart of Chapter 7. The short version: **phones run only one VPN at a time**, so the Aura/Mullvad app and Tailscale are mutually exclusive on a phone.

Option C: Your Pi as a self-hosted exit node (free, but not private)

You can advertise one of your *own* machines as an exit node:

```
sudo tailscale set --advertise-exit-node
# then approve it: admin console -> Machines -> the device ->
#   Edit route settings -> enable "Use as exit node"
```

On Linux you also enable IP forwarding (`net.ipv4.ip_forward=1` and the IPv6 equivalent).

- **What it's good for**: routing through your **home** Pi so you appear to be at home, reaching home-only services, or content tied to your home connection, and, importantly, it's the one exit option that keeps your traffic on your own network where **Pi-hole still applies**.
- **What it's not**: privacy. Traffic exits from **your own IP**, so there's no hiding. And don't use a *phone* as an exit node. It routes in userspace and is slow; the Pi (kernel routing) is fine.

Option D: A self-hosted exit node, made private with Mullvad (*advanced*)

Option C hides nothing, because the Pi exits from your own home IP. You can fix that by *chaining*: run an exit node that itself tunnels out through a standalone Mullvad **WireGuard** config. Your devices reach the node over the tailnet, and the node forwards their traffic out through Mullvad:

you -> Tailscale -> your exit-node box -> Mullvad WireGuard -> internet

This is the only way to use a **standalone** Mullvad subscription *as a Tailscale exit node* without paying for the add-on. The shape of it:

1. Use a machine that is **not your home Pi** if privacy is the goal (a cheap Linux VPS works well). The home Pi only makes sense for “appear at home,” where adding Mullvad would defeat the purpose.
2. On that box, install Tailscale, advertise it as an exit node (`sudo tailscale set --advertise-exit-node`, then approve it in the admin console), and enable IP forwarding.
3. Download a **WireGuard config** from your Mullvad account and bring it up (`wg-quick up <conf>`) so the box’s default route leaves through Mullvad.
4. The sharp edge: Mullvad’s config claims the **entire** default route (`AllowedIPs = 0.0.0.0/0`), which also swallows the return path to your tailnet. You have to keep Tailscale’s range (`100.64.0.0/10`) out of the Mullvad tunnel so your devices can still reach the node. That policy-routing fiddliness is exactly the work the add-on saves you.

Reach for this if you already pay for Mullvad and enjoy the plumbing. For most people the add-on (Option A) buys the same outcome for about \$5/month with none of the routing surgery.

Can you run the VPN *and* Pi-hole? (home vs. the road)

This is the question that trips everyone up, because a privacy VPN and Pi-hole both want to own your **DNS**, and that overlap is where they collide.

- **At home, on a computer, it is mostly a non-issue, because you usually do not need the privacy VPN here.** Pi-hole is already cleaning ads for every device on the network, and you trust your own connection, so the normal state is *Pi-hole on, privacy VPN off*. When you do want to change location or hide your IP for a bit, you flip a standalone Mullvad config on for that session. While it is on, Mullvad carries your DNS too, so for that window you lean on Mullvad’s own built-in ad and tracker blocking instead of Pi-hole, then switch it back off. A standalone Mullvad subscription handles this case on a laptop perfectly well, with no Tailscale involved.
- **Away from home is where it actually bites,** because now you want *both* at once: the privacy VPN (you are on untrusted Wi-Fi) *and* your tailnet and Pi-hole. On a **phone that is impossible with two separate apps**, since a phone runs only one VPN tunnel at a time, so Mullvad’s app and Tailscale cannot both be on. That single limit is the whole reason the add-on exists.

So on the road you have two honest choices:

- **Convenience, the Tailscale Mullvad add-on (Option A).** One tunnel does both: you stay on Tailscale, so the tailnet and Pi-hole DNS keep working, while your traffic exits through Mullvad. Nothing to toggle.
- **Separate and switch, a standalone Mullvad app *plus* Tailscale, picking one in the moment.** Mullvad on when privacy matters most (you give up the homelab and Pi-hole for that stretch), Tailscale on when you need the homelab or Pi-hole (you give up the privacy VPN). It costs nothing beyond the Mullvad sub and stays fully flexible; you just cannot have both at the same instant on a phone.

[Chapter 7](#) is the field manual for living with this on the road.

Recommendation

For a homelab that’s already built on Tailscale and a user who wants privacy *and* wants it to mesh with everything else:

- **For privacy / location changing: Option A (Tailscale’s Mullvad add-on).** It’s the same price ballpark as Aura or a standalone Mullvad sub, but it lives inside the one VPN your devices already run, so it’s the only choice that works cleanly on iPhone alongside the rest of the homelab, and the setting persists across reboots for true “always on.”
- **Keep Option C (Pi as exit node) in your pocket** as a complement, not a replacement: flip to it when you want to appear at home or keep Pi-hole filtering active.
- **A consumer app like Aura (Option B) is fine to keep** as a standalone tool, but understand it won’t integrate with the homelab. It can’t run on the Pi, and on your phone it can’t run at the same time as Tailscale. That trade-off, and how to live with it on the road, is precisely what Chapter 7 unpacks.

i “Always on” in one line

With Option A, `tailscale set --exit-node=<mullvad-node>` (or picking a location in the mobile app) sticks across reboots and reconnects. You set it once per device. If the exit node ever becomes unreachable, traffic *fails closed* rather than leaking out your real connection, which is the behavior you want for an always-on privacy setup.

Recap

- A **VPN** hides your IP and shields your traffic from the local network, but it isn’t anonymity and doesn’t block ads on its own.
- A Tailscale **exit node** is a VPN built from your tailnet; with the **Mullvad add-on**, Mullvad’s servers *are* your exit nodes, inside the app you already run.
- Four options: **A)** Tailscale’s Mullvad add-on (best fit, works on iPhone), **B)** a standalone app like Aura or Mullvad’s own, **C)** your Pi as a free-but-not-private exit node, and **D)** a standalone Mullvad chained behind your own exit node (advanced, no add-on).
- **At home** you rarely need the privacy VPN, so Pi-hole just runs; **on the road** a phone forces you to pick one, which is why the add-on (one tunnel for both) is so convenient.
- **Recommendation:** use Option A for privacy, keep Option C for appear-at-home, and know a consumer VPN’s limits before relying on it.

You now know your VPN options and how to turn each one on. The hard part, what actually happens to your Pi-hole, your homelab access, and your VPN when you walk out the front door, and how to get a setup that doesn’t fight itself, is next, in [Chapter 7](#).